

Motivation

- Cloud-free ASR** — Cloud APIs add latency and privacy risk; all inference runs entirely on-device.
- Edge constraints** — Jetson Orin Nano (4 GB, 20 TOPS) demands model optimization to meet real-time targets.
- Sensor unification** — USB microphone treated as a structured sensor, co-logged with environmental data (BME680).
- SAGE integration** — Transcripts published via REST / WebSocket to SAGE testbed nodes with zero broker dependency.

Benchmark Results

Model	Accuracy %	Infer. ms	GPU %	CPU %
tiny.en	100.0	2118	0.0	62.2
CPU				
base.en	97.3	4045	0.3	64.1
CPU				
tiny.en	37.8	1267	28.7	21.5
GPU				
base.en	35.1	1544	39.6	19.7
GPU				

GPU accuracy note: CPU achieves 97–100% on the reference transcript. GPU reports lower token-match (~35–38%), GPU outputs showed lower transcript similarity than CPU outputs, possibly due to differences in decoding or CUDA inference behavior in whisper.cpp. Further investigation is ongoing. GPU is 1.6–2.6× faster.

Limitations

- GPU decoder output** — whisper.cpp CUDA path yields low token-match vs CPU; root cause under investigation.
- Fixed 30 s windows** — No streaming VAD; transcript appears only after each full chunk.
- English-only tested** — Multilingual models increase VRAM and inference time significantly.
- Noise sensitivity** — WER degrades in high-ambient-noise without pre-filtering stage.

Future Work

- faster-whisper backend** — Switch to CTranslate2 to resolve GPU decoder accuracy gap and compare INT8 gains directly.
- Streaming VAD** — Silero-VAD for utterance-level chunking, targeting < 300 ms perceived latency.
- Multi-sensor fusion** — Correlate acoustic events with BME680 thermal spikes for anomaly detection.

System Architecture



Key Findings

- Near Real-time transcription** — tiny.en GPU completes inference in 1267 ms per 30 s chunk — within capture window.
- CPU accuracy is robust** — 97–100% word accuracy on CPU; GPU output requires decoder investigation.
- Low CPU footprint** — GPU mode uses only 19–22% CPU, leaving headroom for concurrent sensor tasks.
- Zero cloud dependency** — Transcripts exposed via sage.evl.uic.edu REST/WebSocket, no broker required.

1267ms

Fastest GPU inference

tiny.en · 5-run avg

100%

Best accuracy

tiny.en CPU mode

2.5×

GPU speedup vs CPU

base.en model

16kHz

Audio sample rate

mono PCM

Data Flow

1	Audio capture	continuous
	PyAudio reads 16 kHz mono stream in 30 s sliding windows from USB mic	
2	Preprocessing	< 50 ms
	Noise reduction → resample → Log-Mel STFT (80 ch) → float32 tensor	
		of total
3	Whisper inference (GPU · tiny.en)	1267 ms avg
	whisper.cpp CUDA — encoder (CNN + Transformer) + decoder (beam search width 5) — reported as single end-to-end Whisper time by benchmark	
		~95% of total
3	Whisper inference (GPU · base.en)	1544 ms avg
	Larger model — same CUDA pipeline, higher GPU utilization (39.6% vs 28.7%)	
		~97% of total
3	Whisper inference (CPU · tiny.en)	2118 ms avg
	CPU-only inference — 1.67× slower than GPU tiny.en; 100% accuracy	
		vs base.en CPU max
4	Persist + broadcast	< 10 ms
	Write transcript + UTC timestamp to SQLite; WebSocket push to subscribers simultaneously	
		< 1% of total
GPU TOTAL (tiny.en)		~1.3 s · real-time ✓
CPU TOTAL (base.en)		~4.1 s · near real-time

References & Acknowledgements

[1] P. Beckman et al., "SAGE: A Software-Defined Sensor Network," IEEE IPDPS Workshops, 2020.

[2] A. Radford et al., "Robust Speech Recognition via Large-Scale Weak Supervision," arXiv:2212.04356, 2022.

[3] OpenAI, "Whisper: Robust Speech Recognition via Large-Scale Weak Supervision," GitHub repository, 2023.

[4] NVIDIA, "Jetson AGX Orin Developer Kit," 2022.

[5] Georgi Gerganov, "whisper.cpp: Port of OpenAI's Whisper model in C/C++," GitHub repository, 2024.

[6] Chart.js Contributors, chartjs.org, 2024.

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